

	CoMD	HPCG	SPARTA	Athena++	CLAMR	PRK_Stencil
A. Loop progress counter	☑/☑	☑/☑	☑/☑	☑/☑	☑/☑	☑/☑
B. Primary mutating arrays	☑/☑	☑/☑	☑/☑	☑/☑	☑/☑	☑/☑
C. Per-rank dynamic counts	☑/☑	☑/☑	☑/☑	☑/☑	☑/☑	—
D. Sim time / dt	☐/☐	☐/☐	☑/☑	☑/☑	☑/☑	☐/☐
E. Conserved/diagnostic scalars	☑/☑	☐/☐	☑/☑	☐/☐	☑/☑	☐/☐
F. Wall-time accumulators	☐/☐	☑/☑	☑/☑	☐/☐	☑/☑	☐/☐
G. RNG state	—	—	☑/☑	—	—	—
H. Solver tunables (defensive)	—	☑/☐	—	—	—	—
I. Geometry / decomposition	☐/☑	☐/☐	☐/☑	☑/☑	☐/☑	☐/☐
J. Species / material tables	☐/☑	—	☐/☑	—	—	—
K. AMR mesh topology	—	—	—	☑/☑	☐/☐	—
L. Sparse matrix / multigrid	—	☐/☐	—	—	—	—
M. Halo / ghost / MPI buffers	☐/☐	—	☐/☐	☐/☐	☐/☐	☐/☐
N. Stdout / harness mirror	☐/☐	☐/☐	☑/☐	☐/☐	☐/☐	☐/☐
O. Restart-format header	☐/☐	☐/☐	☑/☑	☑/☑	☑/☐	☐/☐

☑/☑ Both save
☑/☐ LLM only
☐/☑ Reference only
☐/☐ Both skip
— Not applicable